

Sample Queuing Model Questions and Answers

Question:

A call center with one operator has an incoming call rate of three calls per hour. On the average, the operator spends 15 minutes for each call. Answer the following questions:

- How busy is the operator?
- How likely is it that a customer will have to wait on hold?
- How long can we expect customers to have to wait on hold?
- What arrival rate can a single operator reasonably handle without causing excessive waiting (less than 5 minutes)?

Question:

Traffic to a message switching center for one of the outgoing communication lines arrive in a random pattern at an average rate of 240 messages per minute. The line has a transmission rate of 800 characters per second. The message length distribution (including control characters) is approximately exponential with an average length of 176 characters. Calculate the following principal statistical measures of system performance, assuming that a very large number of message buffers are provided:

- Average number of messages in the system
- Average number of messages in the queue waiting to be transmitted.
- Average time a message spends in the system.
- Average time a message waits for transmission
- Probability that 10 or more messages are waiting to be transmitted (not including the message being transmitted)

Answer: This is an M/M/1 model

$E[s] = \text{Average Message Length} / \text{Line Speed} = \{176 \text{ char/message}\} / \{800 \text{ char/sec}\} = 0.22 \text{ sec/message}$

$\mu = 1 / 0.22 \{ \text{message} / \text{sec} \} = 4.55 \text{ message} / \text{sec}$

$\lambda = 240 \text{ message} / \text{min} = 4 \text{ message} / \text{sec}$

$\rho = \lambda / \mu = 0.88$

- a. $L = \rho / (1 - \rho) = 7.33$ (messages)
- b. $L_q = \rho^2 / (1 - \rho) = 6.45$ (messages)
- c. $W = E[s] / (1 - \rho) = 1.83$ (sec)
- d. $W_q = \rho \times E[s] / (1 - \rho) = 1.61$ (sec)
- e. $P \{11 \text{ or more messages in the system}\} = P\{N \geq 11\} = \rho^{11} = 0.245$

Question:

A branch office of a large engineering firm has one on-line terminal that is connected to a central computer system during the normal eight-hour working day. Engineers, who work throughout the city, drive to the branch office to use the terminal to make routine calculations. Statistics collected over a period of time indicate that the arrival pattern of people at the branch office to use the terminal has a Poisson (random) distribution, with a mean of 10 people coming to use the terminal each day. The distribution of time spent by an engineer at a terminal is exponential, with a mean of 30 minutes. The branch office receives complains from the staff about the terminal service. It is reported that individuals often wait over an hour to use the terminal and it rarely takes less than an hour and a half in the office to complete a few calculations. The manager is puzzled because the statistics show that the terminal is in use only 5 hours out of 8, on the average. This level of utilization would not seem to justify the acquisition of another terminal. What insight can queuing theory provide?

Answer: This is an M/M/1 model

$$\lambda = 10 \text{ person / day}$$

$$= \{10 \text{ person / day}\} \times \{1 \text{ day / 8hr}\} \times \{1\text{hr} / 60 \text{ min}\} = 10 \text{ person / 480 min} = 1 \text{ person / 48 min}$$

$$\text{In other words: } \lambda = 1 / 48 \text{ (person / min)}$$

$$E[s] = 30 \text{ minutes : 1 person}$$

$$\text{Thus: } \mu = 1 / 30 \text{ (person / min)}$$

In summary:

- Arrival Rate = $\lambda = 1 / 48$ (customer / min)
- Server Utilization = $\rho = \frac{\lambda}{\mu} = \frac{\frac{1}{48}}{\frac{1}{30}} = \frac{30}{48} = \frac{5}{8} = 0.625$
- Probability of 2 or more customers in system = $P\{N \geq 2\} = \rho^2 = 0.391$
- Mean number in the system = $L = \rho / (1 - \rho) = 1.667$
- Mean time a customer spends in the system = $W = 1 / (\mu - \lambda) = 80 \text{ (min)}$
- Mean number of customers in queue = $L_q = \rho^2 / (1 - \rho) = 1.04$
- Mean time in queue = $W_q = \lambda / (\mu (\mu - \lambda)) = 50 \text{ (min)}$

Question:

Assume that you have a printer that can print an average file in two minutes. Every two and a half minutes a user sends another file to the printer.

- How long does it take before a user can get their output?
- To speed things up you can buy another printer that is exactly the same as the one you have. How long will it take for a user to get their files printed if you had two identical printers?
- Another solution is to replace the existing printer with one that can print a file in an average of one minute. How long does it take for a user to get their output with the faster printer?

Answer: (a) is M/M/1 model, (b) is M/M/2 model, and (c) is M/M/1 model

$$\lambda = 1/2.5 = 0.4 \text{ (file/min)}$$

For slow printer: $E[s] = 2 \text{ (min)}$. Thus: $\mu = 0.5 \text{ (file / min)}$

- $W = 1/(\mu - \lambda) = 1/0.1 = 10 \text{ (min)}$
- $W = Wq + (1/\mu)$
 $\rho = \lambda/(c\mu) = 0.4/(2 \times 0.5) = 0.4$

$$\text{From M/M/c formula: } P_0 = \left(\sum_{n=0}^{c-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^c}{c!} \cdot \frac{1}{1 - (\lambda/(c\mu))} \right)^{-1}$$

$$\text{Thus: } P_0 = \left(1 + \frac{(0.4/0.5)^1}{1!} + \frac{(0.4/0.5)^2}{2!} \times \frac{1}{1 - (0.4/(2 \times 0.5))} \right)^{-1} = \frac{1}{1 + 0.8 + \frac{0.64}{2} \times \frac{1}{0.6}} = \frac{1}{2.33} \approx 0.43$$

$$L_q = \frac{(\lambda/\mu)^c \rho}{c! (1 - \rho)^2} P_0 = \frac{(\lambda/\mu)^2 \rho}{2! (1 - \rho)^2} P_0 = \frac{(0.4/0.5)^2 \times 0.4}{2 \times (1 - 0.4)^2} \times 0.43 = 0.15$$

$$\text{And, } Wq = Lq/\lambda = 0.15/0.4 = 0.375$$

$$\text{Thus: } W = 0.375 + (1/0.5) = 2.375 \text{ (min)}$$

- For this question, λ remains unchanged. However: $\mu = 1 \text{ (file/minute)}$
 Thus: $W = 1/(\mu - \lambda) = 1/0.6 = 1.67 \text{ (min)}$

It is quite obvious that replacing the existing printer with a faster one will lead to shorter time for a user to get their output.

Question:

The unisex barbershop can hold only three customers, one in service and two waiting. Additional customers are turned away when the system is full. On the average, a customer arrives at the barbershop for every half an hour and the barber usually spends 20 minutes to provide service for one customer.

- What is the waiting time of each customer?
- How busy is the barber?
- What is the rate of customer being turned away due to there is no available space?

Answer: This is an M/M/C/K model (C = 1 and K = 3)

$\lambda = 1 \text{ customer} / 0.5 \text{ hour} = 2 \text{ (customer/hour)}$

$E[s] = 20 \text{ minutes} : 1 \text{ customer}$

Thus: $\mu = 3 \text{ (customer / hour)}$

$\rho = \lambda/\mu = 2/3$

$$P_0 = \frac{1-\rho}{1-\rho^{K+1}} = \frac{1-2/3}{1-(2/3)^{3+1}} = \frac{1/3}{1-16/81} = 0.415$$

As $P_n = \rho^n P_0$, then: $P_1 = 0.277$, $P_2 = 0.185$, and $P_3 = 0.123$

And: $\lambda_e = \lambda(1-P_K) = 2 \times (1 - 0.123) = 1.754 \text{ (customer/hour)}$

$$L = \frac{\rho}{1-\rho} - \frac{(K+1)\rho^{K+1}}{1-\rho^{K+1}} = \frac{2/3}{1-2/3} - \frac{(3+1)(\frac{2}{3})^{3+1}}{1-(\frac{2}{3})^{3+1}} = 2 - \frac{4 \times (\frac{16}{81})}{1-\frac{16}{81}} = 2 - \frac{64}{65} = 1.015 \text{ (customer)}$$

$$L_q = L - (1 - P_0) = 1 - (1 - 0.415) = 0.43 \text{ (customer)}$$

- The waiting time of each customer = $W_q = L_q / \lambda_e = 0.43/1.754 = 0.245 \text{ (hour)}$ or about 15 minutes
- Utilization = $\lambda_e / \mu = 1.754/3 = 0.585$
- The loss rate of this model = $\lambda - \lambda_e = 2 - 1.754 = 0.246 \text{ (customer / hour)}$